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Utilizing Computational Analysis as an Aid in Understanding Overlapping Gene Expression in UMAP Clusters

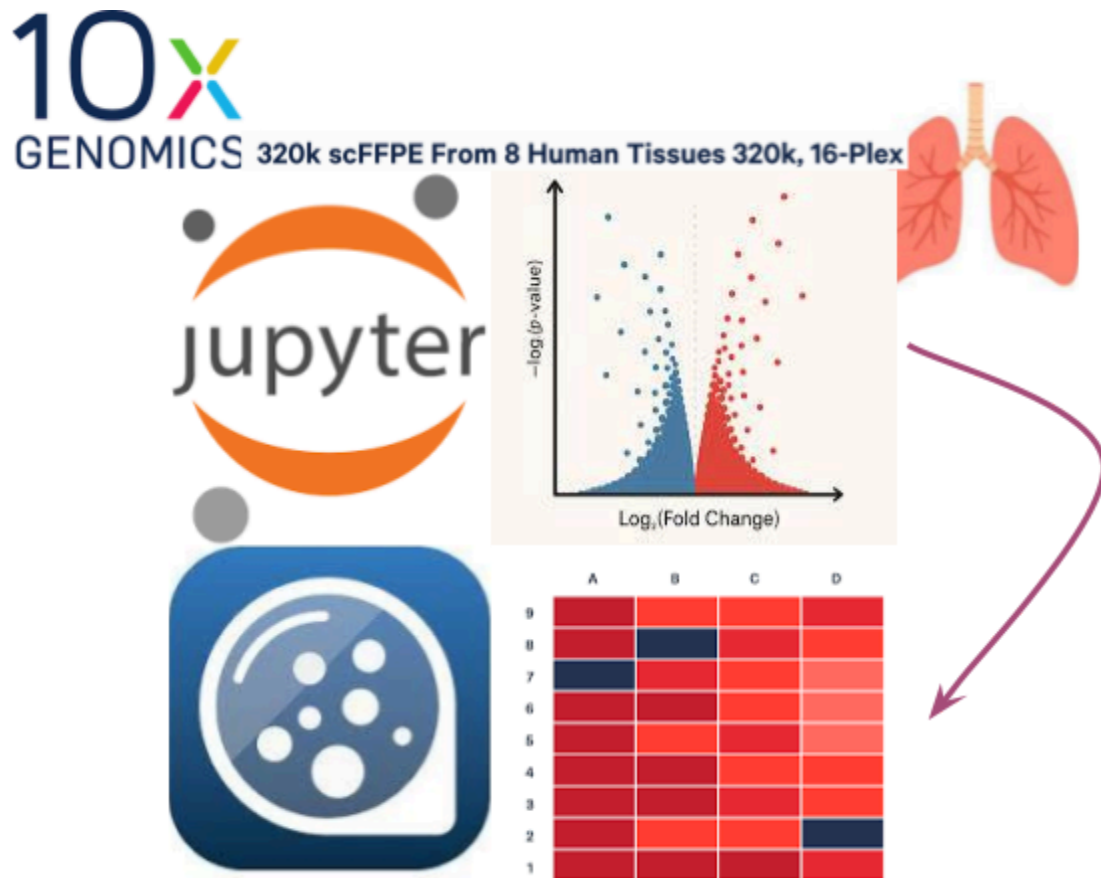


Figure 1. Simple representation of the tools used in the final project exploring the use of Jupyter Notebook and code as a tool in understanding clustering of cells and their gene expression in single-cell data from a lung cancer sample.

Images: (Top left to bottom right) 10xGenomics logo and title of dataset used from 10xGenomics. Lung image from [Adobe Stock](#). Jupyter logo from Jupyter/Anaconda. Volcano plot illustration from [Facebook](#). Loupe Browser logo from Loupe. Heatmap illustration from [Data Viz Project](#).

INTRODUCTION

This project looks at a large dataset of about 69,000 cells from a human lung cancer sample provided by 10x Genomics. The main goal is to sort these cells into groups and understand the specific genes that make them different and whether those genes are shared by others.

To do this, I used two main tools: Loupe Browser and Jupyter Notebook. Loupe Browser was used to create a map of the cells (called a UMAP) and identify 26 different clusters. I then used Jupyter to write code for "volcano plots," which are special charts that show which genes are the most important for each group. By using these tools together, I was able to organize the cells into three big categories: tumor cells, immune cells, and fibroblasts. The three categories were chosen based on the typical cells present in cancer microenvironments. Tumor and immune cells have tons of crosstalk (Wang et Al, 2019), as immune cells like T-cells and B-cells work against the tumor, or can even be low in numbers in cancer samples. Cancer-associated Fibroblasts (CAFs) are also important in the tumor microenvironment (TME). CAFs are in overwhelming abundance in the TME and support tumor cells (Yang et. al, 2023).

From this, I expect to find those three categories of clusters to be relatively close to each other, as UMAP is a dimensionality reduction tool that preserves local and global structure.

Additionally, the highly upregulated/downregulated genes presented by the Jupyter notebook are expected to show high expression in all the respective categories' clusters.

METHODS

Data

Data was downloaded from the 10xGenomics public dataset. Downloaded was a 'Gene Expression' Cloupe file of lung cancer tissue was downloaded from a larger dataset looking at 8 total cancer tissues from varying organs ('320k scFFPE From 8 Human Tissues 320k, 16-Plex'). Preservation of the tissue was an FFPE block obtained from Avaden Biosciences from a human lung. The single-cell sample contained about 69,000 cells.

Loupe Browser 9

Once the data was downloaded, it was analysed using Loupe Browser 9, an application downloaded onto a computer. The general visualization of the data was a UMAP. 26 clusters were represented in the dimensional reduction technique.

Identifying cell types + groups

To identify the cell types for the 26 clusters, the 'features table' provided by Loupe displayed the upregulated genes for each cluster along with their respective Log2 Fold Change and P-value. Screenshots of the top 10 upregulated genes and their corresponding L2FC and P-value for each cluster were uploaded onto Google Gemini, an AI tool. The following prompt was given: "Can you identify the cluster's cell type based on the top upregulated genes? The data comes from a lung cancer sample."

From the 26 newly identified clusters, they were grouped into 3 broader categories of tumor cells, immune cells, and Fibroblasts.

Volcano Plots

Jupyter notebooks were used to code volcano plots. A CSV file of 'all gene expression' was downloaded from the Loupe browser. Both the x-axis and y-axis were provided by the data; only the y-axis had to be transformed. With p-values being small, a $-\log_{10}(\text{p-value})$ transformation was done to the data. Individual volcano plots were made for the clusters, and combined volcano plots for the 3 categories. Top genes of importance for both downregulated and upregulated were coded to be listed.

Heat Maps

Using the top genes listed from the Jupyter notebook, three (tumor, immune, fibroblast) separate heat maps and UMAP visualizations were made. In the individual heat maps, a gene per cluster of the category was selected to view its expression in the other clusters a part of the category. Heat maps were made in Loupe Browser 9, under the features tab.

Violin Plots

Using the genes selected for the heat maps, violin plots were made to show the expression distribution in each cluster. The plots were made using Loupe Browser.

RESULTS AND CONCLUSIONS

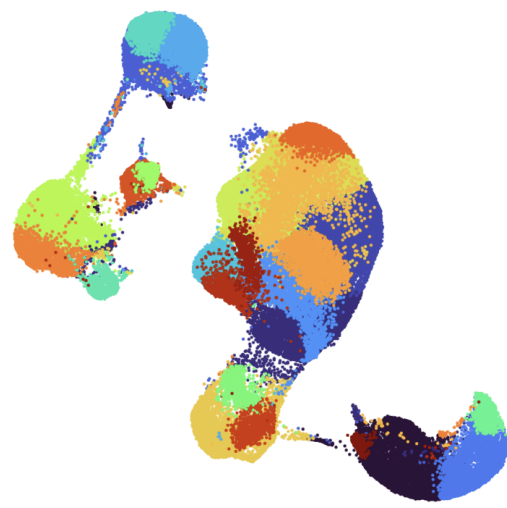


Figure 2. UMAP visualization of all 26 clusters from lung cancer single-cell data. Each point represents an individual cell.

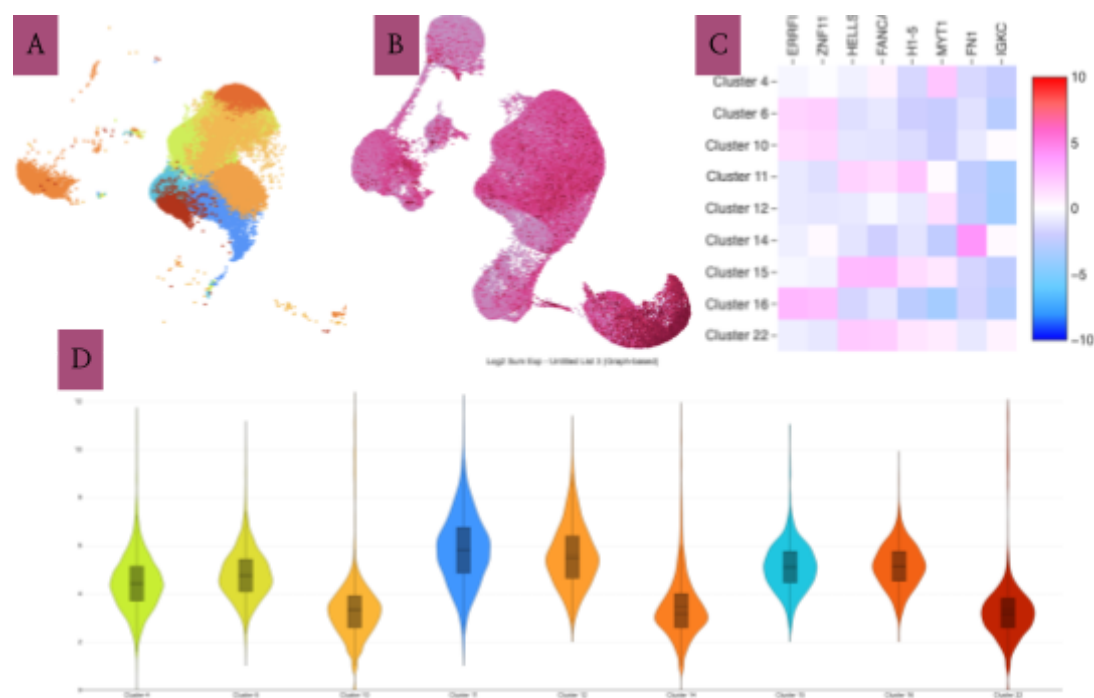


Figure 3. Multipanel figure showing “tumor” clusters’ data created from Loupe Browser 9. A) UMAP visualization of clusters 4, 6, 10, 11, 12, 14, 15, 16, and 22. B) UMAP visualization of a heat map of gene expression for the top genes of the “tumor” category. Darker colors represent higher expression. C) Heat map of top genes selected and their expression in “tumor” clusters.

D) Violin plots of the top genes from the “tumor” category, where the x-axis is the clusters, and the y-axis is Log_2FC .

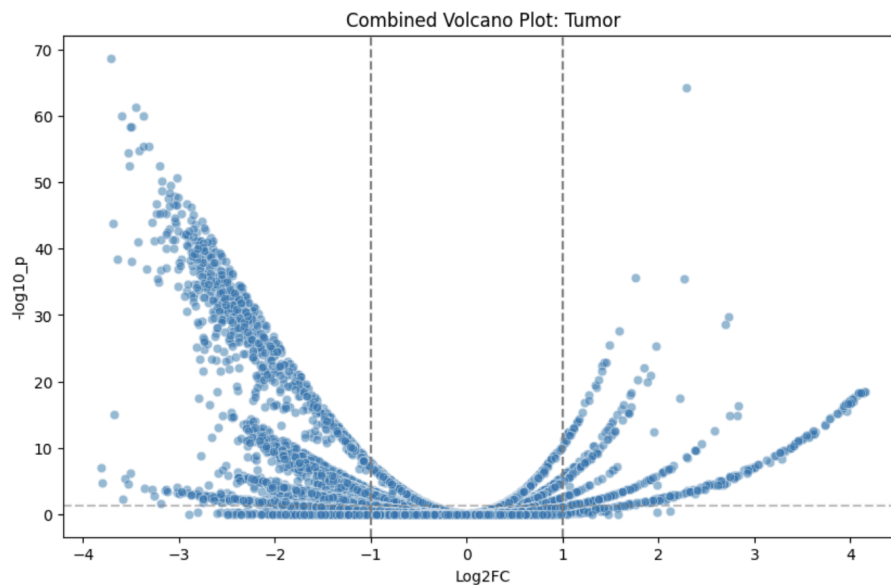


Figure 4. Volcano Plot of the combined “tumor” clusters. Gray dashed lines represent the thresholds. A negative log_2FC indicates downregulated genes, positive being upregulated.

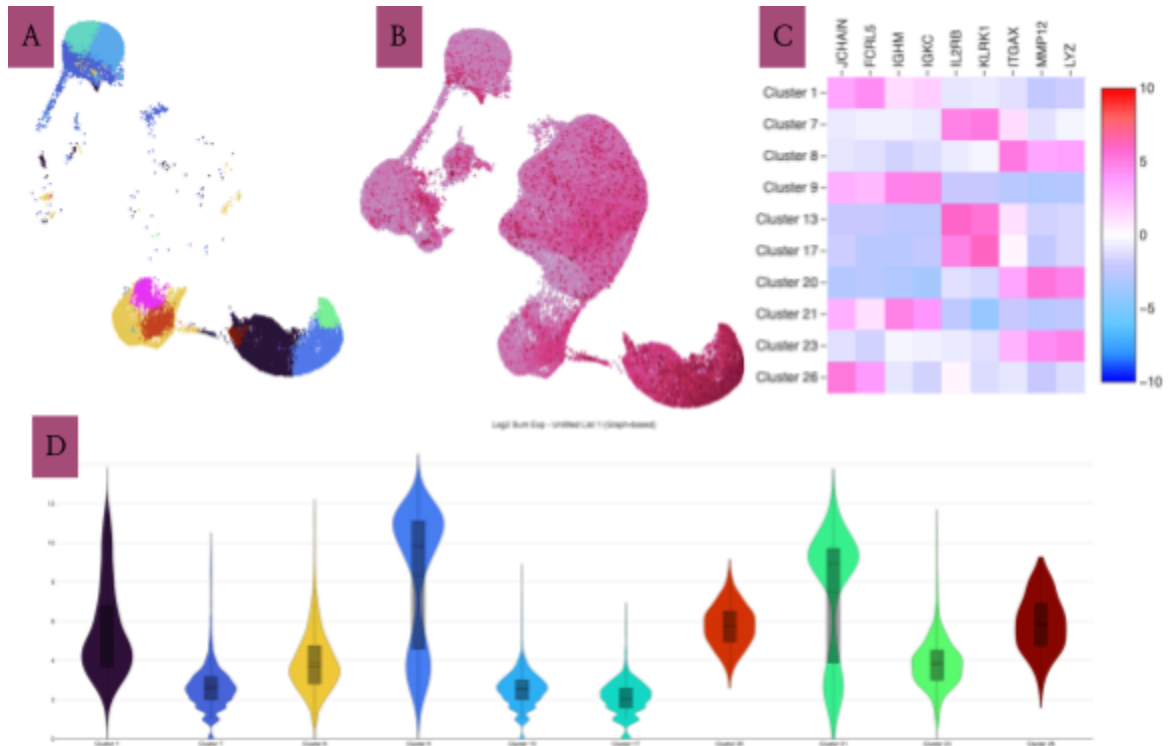


Figure 5. Multipanel figure showing “immune” clusters’ data created from Loupe Browser 9. A) UMAP visualization of clusters 1, 7, 8, 9, 13, 17, 20, 21, 23, and 26 . B) UMAP visualization of a heat map of gene expression for the top genes of the “immune” category. Darker colors represent higher expression. C) Heat map of top genes selected and their expression in “immune” clusters. D) Violin plots of the top genes from the “immune” category of clusters.

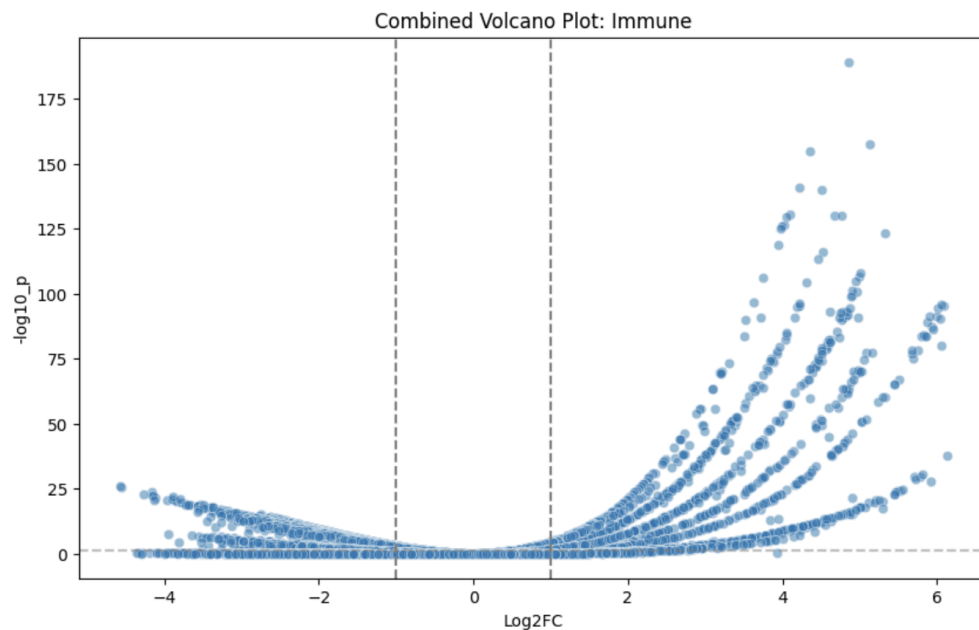


Figure 6. Volcano Plot of the combined “immune” clusters. Gray dashed lines represent the thresholds. A positive Log2FC is upregulation of genes.

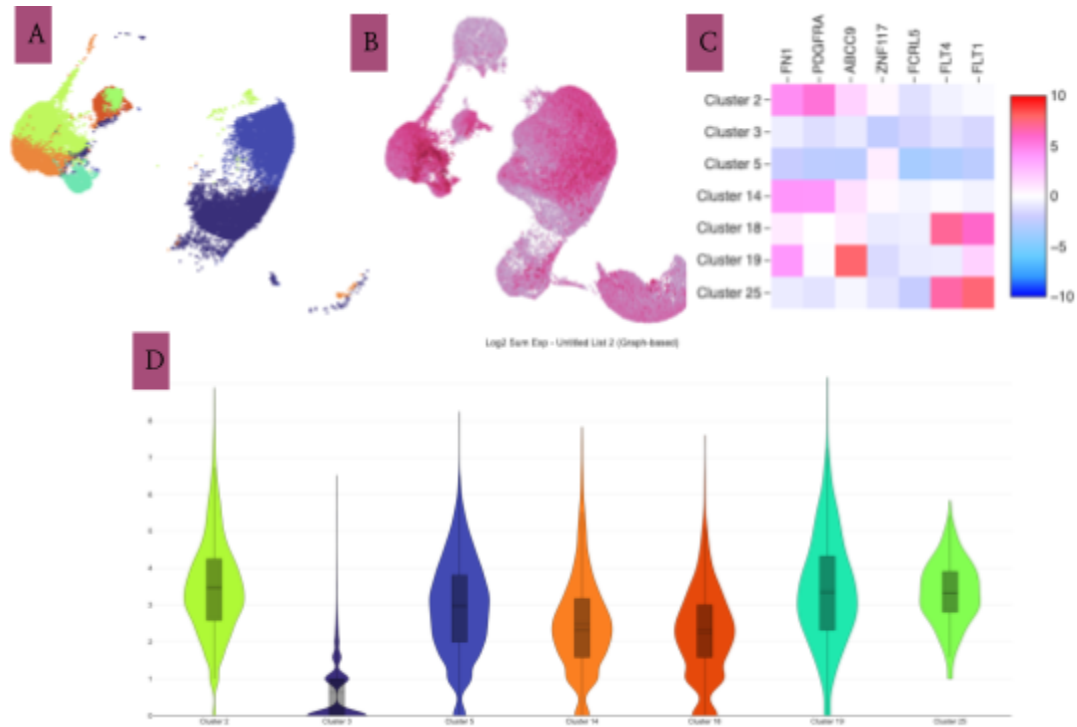


Figure 7. Multipanel figure showing “fibroblast” clusters’ data created from Loupe Browser 9.

A) UMAP visualization of clusters 2, 3, 5, 14, 18, 19, and 25

. B) UMAP visualization of a heat map of gene expression for the top genes of the “fibroblast” category. Darker colors represent higher expression. C) Heat map of top genes selected and their expression in “fibroblast” clusters. D) Violin plots of the top genes from the “immune” category of clusters.

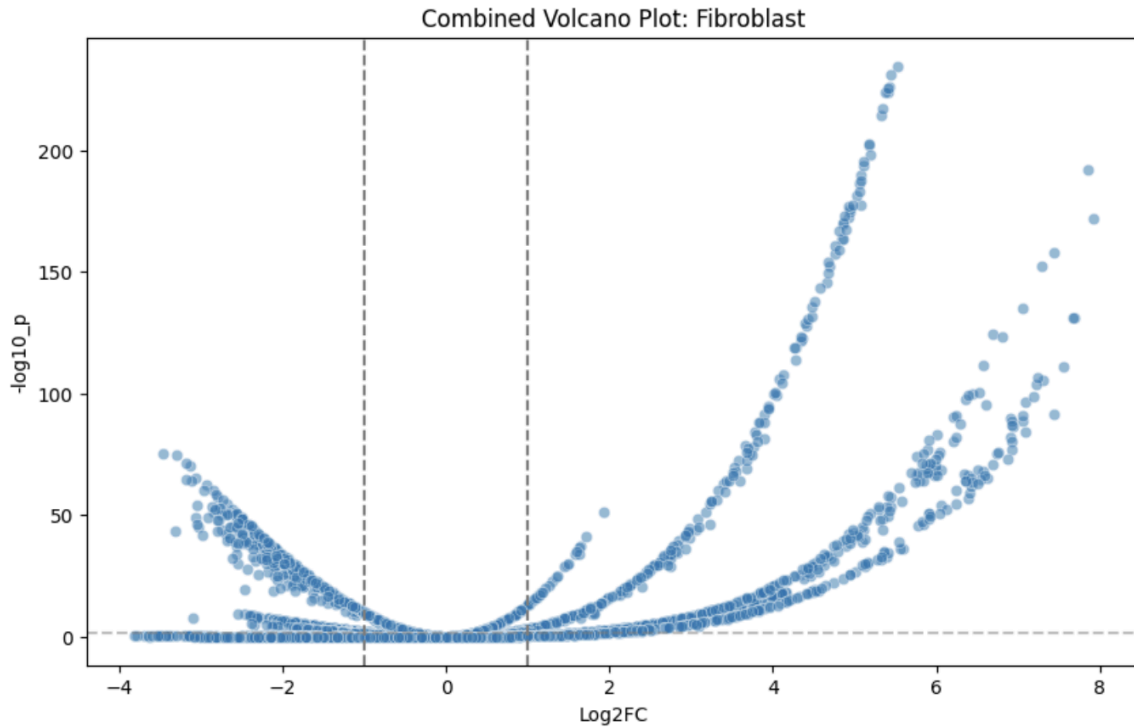


Figure 8. Volcano Plot of the combined “fibroblast” clusters. Gray dashed lines represent the thresholds. A positive Log_2FC is upregulation of genes.

This project shows how computational tools can help make sense of complex single-cell RNA sequencing data that may be hard to digest. By using Loupe Browser and Jupyter Notebook, the cells were grouped into three main categories: tumor cells, immune cells, and fibroblasts. The results generally supported the expectation that cells in the same category would cluster together (Figure 2A, 4A, and 6A). However, it appears that the gene expression of the top genes was not only shared among the categories, but also among the entirety of the 26 clusters. This is supported by the UMAP heatmap representation (Figures 2B, 4B, and 6B), where throughout the three categories, high expression of the category's top genes was seen throughout the other two categories as well.

The volcano plots, heat maps, and violin plots all helped identify genes that were strongly upregulated or downregulated in each category. These gene patterns were consistent with known roles of tumor cells, immune cells, and fibroblasts, supporting the accuracy of the groupings.

One limitation is the use of an AI tool to identify cell types, which may lead to some misclassification. As experienced in previous projects, Google Gemini can change its response easily. However, since there are tons of genes in the dataset, many of which the average Biology major may not know, it was a helpful tool in classification. Another limitation may have been the code. The volcano plots were made with a very simple code and do not look like the typical volcano plot. Additionally, I am unaware of how Jupyter decided on the “top genes.”

Overall, this project demonstrates that combining visualization and coding tools could be effective for analyzing single-cell data, while also highlighting the complexity of the tumor microenvironment.

AI Disclaimer: AI was used for the classification of cell types.

Sources:

Wang, Jing et al. “Crosstalk between cancer and immune cells: Role of tumor-associated macrophages in the tumor microenvironment.” *Cancer Medicine* vol. 8,10 (2019): 4709-4721. doi:10.1002/cam4.2327

Yang, D., Liu, J., Qian, H. et al. Cancer-associated fibroblasts: from basic science to anticancer therapy. *Exp Mol Med* 55, 1322–1332 (2023).
<https://doi.org/10.1038/s12276-023-01013-0>